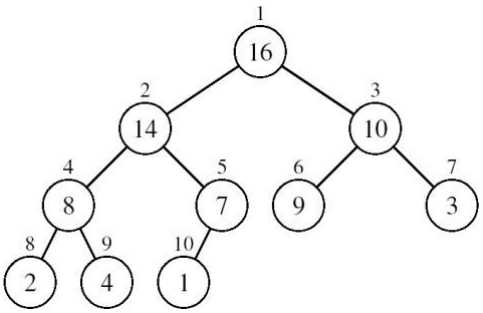
MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY
Data Structures Lab

Assignment 10

1. Write a C program to construct the binary tree if preorder and in-order traversal is given. Print the elements in sorted order.
   Example: Inorder Traversal: { 4, 2, 1, 7, 5, 8, 3, 6 }
   Preorder Traversal: { 1, 2, 4, 3, 5, 7, 8, 6 }
   Construct the Binary tree and also find postorder traversal.

2. Write a C program to construct the binary tree if postorder and in-order traversal is given. Print the elements in sorted order.
   Example: inorder = {4, 2, 1, 7, 5, 8, 3, 6}
   postorder = {4, 2, 7, 8, 5, 6, 3, 1}

3. Write a C program to implement heap sort on a given tree:



4. WAP to implement Binary Search Tree (BST).
   - Insert a node into the BST.
   - Delete a node from the BST.
   - Display the preorder traversal of the BST.
   - Display the inorder traversal of the BST.
   - Display the postorder traversal of the BST.

5. WAP to implement right skewed BST and display the inorder traversal of the tree.

6. WAP to implement left skewed BST and display the postorder traversal of the tree.

7. WAP to implement the following:
   - Count the number of nodes in a binary tree.
   - Count the number of leaf nodes in a binary tree.
   - Count the number of non-leaf nodes in a binary tree.
   - Return the height of the binary tree.
   - Check whether the tree is a strict binary tree or not.
   - Check whether the two trees are equal or not.